16 Conclusion

16.1 General

16.1.1.1 An EIA Report has been prepared for Yuen Long South (YLS) development in accordance with the requirements given in the EIA Study Brief (ESB-246/2012) and the EIAO-TM. Environmental assessment of the potential environmental impacts associated with the construction and operation of the Project has been conducted. Environmental issues covered in this EIA include:

- Air quality impact;
- Noise impact;
- Water quality impact;
- Sewage and sewerage treatment implications;
- Ecology impact;
- Fisheries impact;
- Landscape and visual impacts;
- Waste management implications;
- Land contamination impact; and
- Cultural Heritage impact.

16.2 Air Quality

16.2.1 Construction Phase

16.2.1.1 Potential construction dust impact would be generated from site clearance, ground excavation, site formation, etc. during the construction phase. Quantitative construction dust impact assessment have been conducted. Results have concluded that there will not be any adverse residual air quality impact during construction phase given frequent watering on all works area once per hour during working hours.

16.2.2 Operational Phase

16.2.2.1 Quantitative operational air quality assessment has been conducted, taking into account the vehicular emission impact associated with the Project and existing road networks, and industrial emission in the vicinity of the Project. It is concluded that the predicted cumulative air quality impacts on all air sensitive receivers would comply with the AQOs during the operational phase of the Project.

16.2.3 Odour Issues during Operation Phase

16.2.3.1 Quantitative odour impact assessment has been conducted, taking into account the proposed STW and existing livestock farm to be retained under the RODP. It is concluded that, during full commissioning of the Project, the predicted cumulative

odour concentrations on all planned air sensitive receivers would comply with the criteria during the operational phase of the Project. Nevertheless, exceedance is predicted at some existing ASRs.

16.2.3.2 Given the fact that, without the Project in place, existing sensitive receivers are subject to an even worse odour impact as they are exposed to other nearby livestock farm. Moreover, a total of five existing livestock farms within the Project site will be removed under this Project which would reduce the total odour emission in the area and improve the odour conditions in future from existing situation.

16.2.3.3 During the interim period, there is also a general improvement in odour conditions, in terms of maximum odour concentration and/or percentage time of the year, at all of the existing ASRs. Predicted odour concentrations at the planned ASRs in Phase 1 would also comply with the criterion.

16.3 Noise

16.3.1 Construction Phase

16.3.1.1 Construction noise associated with the use of PME for different stages of construction has been conducted. With the implementation of practical mitigation measures including good site management practices, use of movable noise barrier, full enclosure and retractable barrier, use of "quiet" plant and working method, construction noise impacts at all of the neighboring residential noise sensitive uses would be controlled to acceptable levels. For educational institutions, the construction noise impacts during school normal and examination periods would also comply with the relevant criteria.

16.3.2 Operation Phase

16.3.2.1 Operational road traffic noise impact on planned and existing noise sensitive uses within the PDA have been predicted for the assessment year. Results indicate that the noise impacts can be mitigated by a combination of noise mitigation measures including 1) absorptive vertical barriers and cantilevered noise barriers along some sections of Project Roads and Kung Um Road; 2) LNRS on some road sections; 3) semi-enclosures or full enclosures at primary distributor roads at TYST Interchange and Kung Um Road; 4) nullah features / barriers along some sections of Yuen Long Nullah (Kiu Hing Road); 5) provision of acoustic windows for some planned public and private housing and 6) alternative building orientation for some planned schools are proposed to alleviate adverse traffic noise impact on the affected NSRs. These mitigation will also ensure that the noise levels caused by project roads are within the respective noise criteria.

16.3.2.2 For the existing road sections such as Kung Um Road, Sha Tseng Road and Tong Yan San Tsuen Road, LNRS would be considered to address the existing noise impacts from existing roads. Different materials for LNRS are being reviewed and tested by the Government for application in Hong Kong. Environmental reviews will be conducted at the later design stage to review the proposed noise mitigation measures taking into account the latest design standard at that time for the application of the LNRS materials. Similar considerations are also applied for planned roads.

16.3.2.3 The proposed at-receiver mitigation measures i.e. the provision of acoustic windows for some planned public and private housing are subject to further study by HKHA and future private developers respectively. HKHA and future developers can further

explore alternative options which can achieve corresponding traffic noise reduction during the detailed design stage.

16.3.2.4 There is scope for adoption of alternative noise mitigation measures, for example, by means of alternative layout and design of individual developments at the detailed design stage. Noise impact assessment at the planned residential sites is proposed to be conducted by future developers at the detailed design stage to study whether the future development layout would avoid exposing to excessive road traffic noise levels so as to minimise the scale and extent of the proposed noise mitigation measures such as noise barriers and special building design. The requirement of noise impact assessment would be included in the land lease condition or planning briefs of the residential sites.

16.3.2.5 In addition, Environmental Assessment Study (EAS) for the planned public housings will be conducted by Hong Kong Housing Authority (HKHA) in the detailed design stage according to HKPSG to address environmental impacts and to comply with relevant criteria.

16.3.2.6 Fixed noise sources impact assessment has been conducted. Noise impact from planned fixed plant could be effectively mitigated by implementing noise mitigation measure at source during the detailed design stage. With the adoption of the proposed maximum permissible SWLs for the proposed fixed plant, the impact noise levels at representative NSRs complies with the relevant noise criteria for both existing and planned fixed noise sources. Therefore, adverse fixed noise sources impact on the NSRs is not anticipated.

16.3.2.7 An EFTS is proposed to be introduced to operate within the Project, which may be in the form of rail based mode of transport. For conservative noise assessment, the rail based EFTS was assumed for rail noise impact assessment. Results indicate that the noise impacts on NSRs would comply with the statutory requirement after considering some track enhancement measures such as embedded rail, green track with vegetation, etc. With the proposed of 10m buffer distance, the operation of EFTS would not pose adverse noise impact on the nearby NSRs.

16.4 Water Quality

16.4.1 Construction Phase

16.4.1.1 During the construction phase, with full implementation of the mitigation measures, no substantial impact is anticipated from surface runoff from construction site and sewage generated from construction workforce in construction phase.

16.4.2 Operation Phase

16.4.2.1 During the operational phase, a new STW will be proposed and designed to achieve a total treatment capacity of 24,000m^3/day in ADWF and up to tertiary level treatment standard to enable reusing of reclaimed water while the rest of sewage is suggested to be treated by secondary plus treatment (with UV disinfection and 75% nitrogen removal) and discharged to Urmston Road submarine outfall. Among the reclaimed water, most of it will be reused for non-portable uses such as toilet flushing in the PDA and in adjacent proposed development; and only a small amount of it will be discharged to a reedbed adjoining to the proposed YLS STW for further polishing before entering Yuen Long Nullah. As discussed in **Chapter 6,** the Project would not generate a net increase in pollution loading to the receiving water in the Deep

Bay WCZ and North Western WCZ. Hence, the policy "No Net Increase in Pollution Loads Requirement in Deep Bay" would be complied with. Also, the Project would also bring benefits by providing sewerage infrastructure to the existing unsewered areas.

16.4.2.2 Contingency measures such as twin rising mains, standby pump and treatment facilities and dual electricity supply or backup power supply facilities would be adopted for the proposed YLS STW and two SPSs to prevent emergency situation as far as practicable. In the very unlikely case that all these measures fail, the discharge from the proposed YLS STW and two SPSs will be delivered to the nearby proposed reedbed and the nearby channelised Yuen Long Nullah respectively. However, the water quality impact due to emergency discharge is anticipated to be short-term and thus insignificant. For other potential water pollution sources arising from the Project such as maintenance flushing of Reclaimed Water Service Reservoir, wastewater from industrial and commercial activities, etc., no significant water quality impact is anticipated with the implementation of mitigation measures.

16.5 Sewerage and Sewage Treatment Implications

16.5.1.1 Existing and planned sewerage infrastructure within and in the vicinity of the Study Area has been reviewed and information of sewage collection, treatment and disposal arrangements in the YL STW and SW STW sewerage catchments have been collected for formulation of baseline information.

16.5.1.2 The sewage flows and loads generation from the YLS PDA are estimated and the proposed sewage treatment and disposal strategy for YLS are accordingly formulated, which involve discharging the YLS sewage flows by the proposed sewerage system to a new STW in the southern part of the PDA for treatment, followed by the reuse of reclaimed water as non-potable water supply for use by YLS PDA and other existing and future development in vicinity including Long Bin and TKT development and Tin Shui Wai area.

16.5.1.3 Equivalent pollution loading from tertiary TSE in YLS STW (upon offsetting by credit gained from the existing livestock farms) could be discharged to Yuen Long Nullah adjacent to the STW with strict compliance to the requirement for "No Net Increase in Pollution Loading", while reedbed is provided a further polish to a small amount of TSE that discharge to Yuen Long Nullah and the Deep Bay. The rest of reclaimed water can be conveyed to the existing inlet chamber of the NWNT effluent tunnel at San Wai for discharging to the Urmston Road Submarine Outfall at North Western WCZ.

16.6 Ecology

16.6.1.1 The RODP has preserved all woodland patches of significant sizes in the PDA and avoided encroachment upon the mature woodland behind Shan Ha Tsuen through locating the road junctions away from the preserved woodlands. The proposed Road D1 will unavoidably run through the eastern edge of agricultural land behind Shan Ha Tsuen. However, the integrity of the mosaic of habitats within Shan Ha Tsuen Agricultural Land and linkages with more natural, less disturbed habitats with higher ecological value to the west and south, namely Grassland/Shrubland, Watercourses and Secondary Woodland will still be retained in the RODP. The resulting loss of agricultural land has been minimized and the fragmentation impact on this habitat is considered to be low, as ecological linkages are limited in the eastern portion and these areas are subject to higher human disturbances.

16.6.1.2 The RODP will also retain three lower sections with 15m buffer on each side of the watercourses which have records of the endemic crab *S. zanklon*. Translocation of this species can be conducted to these three streams or the proposed 575m Hillside River Corridor as receptor sites should they occur in those fragmented watercourses in the lowlands.

16.6.1.3 Loss of sections of watercourse which have an ecological value of Low to Moderate or above, or potentially provide suitable habitat for *S. zanklon* and/or other aquatic fauna of conservation importance, are considered to be of Low to Moderate severity. This loss of habitats can be mitigated by the creation of the Hillside River Corridor and the Retention Lake. A translocation programme for *S. zanklon* and/or other aquatic fauna of conservation importance can be implemented, using these new created habitats or retained suitable watercourses as receptor sites.

16.6.1.4 Additional feeding opportunities for breeding ardeids may be provided through the creation of the 0.6ha Retention Lake, the Reedbed and the Hillside River Corridor and through the protection and enhancement of the three retained watercourses with GB(1) buffers and the revitalisation of nullahs.

16.6.1.5 A reedbed with an area of approximate 4ha is proposed to be built at the south of Area 3. Some of the tertiary TSE will be conveyed to the reedbed before being discharged into Yuen Long Nullah can provide ecological enhancement to the area providing foraging and shelter opportunities for a suite of aquatic fauna, and in particular, wetland dependent birds, especially with the inclusion of a freshwater marsh before the outfall.

16.6.1.6 The impacts resulting from loss of hillside secondary woodland (~2.42ha in total) is considered to be of moderate severity and will require appropriate ecological mitigation. On-site woodland compensation appears to be impractical due to limited space within the PDA. An off-site/ near-site for compensatory woodland planting (~12ha) has been proposed to provide a compensation area ratio of at least 1:1 to adequately mitigate the impacts from woodland loss.

16.7 Fisheries

16.7.1.1 A comprehensive review of the status of pond fish culture has been conducted through desktop study and field surveys to establish the fisheries baseline of the Assessment Area, with the assessment of fisheries conducted in accordance with the EIAO-TM requirements.

16.7.1.2 Of the total 51 ponds identified in the Assessment Area, only nine ponds were identified within PDA or associated works footprints. These include five shallow farm ponds for irrigation near Sha Tseng Tsuen in Area 1, a small shallow pond appeared to be actively managed with uncertain use located to the east of Lam Tai East Road in Area 2, two abandoned and overgrown ponds located near the southern end of Kung Um Road, and an abandoned and dry pond located near the southern end of Long Hon Road in Area 1 of the PDA. The original farmland adjacent to Sha Tseng Tsuen will be resumed and converted into "District Open Space" ("DO") under the RODP. Therefore, there will be no demand for irrigation from the five shallow farm ponds after the development. Of these nine ponds identified in the PDA, the abandoned and dry pond in Area 1 will be preserved in the proposed GB zone in the RODP. The remaining eight ponds will be permanently lost due to the Project. As all the eight ponds to be lost within the PDA serving non-fisheries related purposes (of approximately 0.58ha in total) are small in overall area, no significant

direct fisheries impacts are anticipated from loss of the pond areas due to construction of the Project.

16.7.1.3 With the separation distances (~150m) from the nearest boundary of the PDA, no direct or indirect impacts on the two possibly active fishponds are anticipated during the construction and operation phases of the Project. As such, no fisheries mitigation measures are considered necessary. In addition, the Project will provide a STW and associated sewerage facilities to serve the proposed development. It is expected that during the operational phase of the Project, sewage and domestic discharge within the development area will be properly collected and treated as detailed in **Chapters 6 and 7**. As such, no significant indirect impacts are anticipated during the operational phase of the Project.

16.7.1.4 With the full implementation of the good site practices to control sewage discharge and site runoff as detailed in **Chapter 6**, no adverse induced water quality impacts on water sources for potential fish pond sites are anticipated.

16.8 Landscape and Visual

16.8.1.1 Given the rural nature of the Project area, the proposed development of the Project will inevitably result in some landscape and visual impacts during construction and operational phases. It is not possible to fully mitigate all landscape impacts in relation to loss of active agriculture land, mature woodland or tree planting for affected LRs and LCAs in the construction period and early operational stage, mainly as long periods of time are required to sufficiently compensate for the associated impacts. In addition, approximately 63% of exiting trees within the development boundary will be potentially affected. Urban design framework has proposed open spaces and greening along development lots for tree preserved or transplanting. Compensatory planting will be provided at a 1:1 ratio if retained is not practical. After compensatory measures are applied, in combination with landscape enhancement proposed by the RODP, the residual impacts to LRs and LCAs can be reduced to slight to moderate levels eventually.

16.8.1.2 The overall visual character in YLS area would be changed by the proposed development after the construction, while the recommended mitigation measures are not be able to compensate the change in character adequately. However, the overall change that is to occur as a result of development of the Project will eventually bring about positive change. The proposed developments will change the existing brownfield site into a planned new town, also with the enhancement proposals goes further by defining the basic design elements, such as parklands and streetscape in YLS. The transformation will result in strong visual interest and character and improved visual outlook for the majority of VSRs, the overall change would enhance the compatibility with the existing visual character of Yuen Long Town. Moreover, the enhanced planting design and revitalised nullah will help to improve aesthetics and create a green, quality and liveable community for housing supply and improving the existing brownfield environment.

16.8.1.3 Cumulative landscape and visual impact during construction and operation phases are predicted to be generated from other concurrent projects which mainly include the HSK NDA. Given the adverse landscape impacts (such as loss of vegetation, fragmentation of some resources and the fundamental change in character of brownfield sites) are predicted to be similar to the YLS development, no additional landscape impacts are predicted with the implementation of mitigation measures. The adverse visual impacts will not be substantial since both of the proposed

developments would create certain degree of visual amenity enhancement to the local community.

16.8.1.4 On review of the likely residual impacts and possibility to alleviate the impact by operation in year 10 when the mitigation measures are matured and the enhancement proposals are well established, the overall residual landscape and visual impacts would be acceptable with mitigation measures during the construction and operational phases.

16.9 Waste Management Implications

16.9.1 Construction Phase

16.9.1.1 Potential waste management implications from the generation of waste during the construction phase have been evaluated. Measures, including the opportunity for on-site sorting, reusing C&D materials etc., are devised in the construction methodology to minimize the surplus materials to be disposed. Recommendations have been made for implementation by the Contractor during the construction period to minimize waste generation and off-site disposal.

16.9.2 Operational Phase

16.9.2.1 The types of waste that would be generated during the operational phase have been assessed. Recommendations have been made to ensure proper treatment and disposal of these wastes.

16.10 Land Contamination

16.10.1.1 This land contamination assessment examined the potential contaminative land use within the PDA and the works areas for the associated infrastructure. The assessment involved desktop review, site surveys and the proposed environmental site investigation (SI) and their potential impacts to future land use.

16.10.1.2 Majority of the potentially contaminated sites could not be accessed to inspect the site conditions during site walkover at the time of preparing the EIA report and permission could not be obtained from the site owners/ operators to carry out the SI works. As such, this land contamination assessment on the potential land contamination was conducted based on desktop review, review of historical aerial photos and a number of peripheral site surveys.

16.10.1.3 A total of 697 potentially contaminated sites were identified, of which over 90% of the potentially contaminated sites are currently used as open area storage, container storage and warehouse sites. However, open area storage and container storage usually comprise a large portion of areas for goods/ container storage with relatively small portion of potentially contaminating activities such as maintenance of equipment and vehicles and fuel storage. Warehouse sites may not be contaminated if they are used to store general packed goods stocks. The contamination (if any) is therefore expected to be localised if the key types of goods/ stocks stored within the abovementioned landuses are not potential sources of contamination. In addition, the land uses of the remaining identified potentially contaminated sites (less than 10%) are only scattered vehicle maintenance workshop, metal workshop, waste recycling workshops, etc. which are not large scale polluting facilities. This further supports that the contamination (if any) would be localised.

16.10.1.4 The possible COCs identified at the potentially contaminated sites include VOCs, SVOCs, Metals, PCRs, PCBs, etc. These COCs are readily treatable with proven remediation techniques in local remediation experience, e.g. decontamination works at the Cheoy Lee Shipyard. By implementing the recommended remediation works, any contaminated site (s) identified within the PDA could be cleaned up prior to construction / development.

16.10.1.5 The recommended remediation works would not only minimise the health risk to the future occupants arising from the exposure of the contaminated soil and/or groundwater, it would also provide the opportunity to reuse the contaminated materials into useful materials for backfilling, which results in minimising the amount of waste disposing into the depleting landfill in Hong Kong and achieving a more sustainable development.

16.10.1.6 Since the potentially contaminated sites are located in private land lots, SI is unlikely to be carried out at this stage. In addition, as the sites are still in operation, it is considered not suitable to carry out the SI at this stage as there may be change in land use prior to construction for both potentially contaminated sites and other surveyed sites. In view of this, further site visits to these potentially contaminated sites are proposed once future development of these sites are confirmed and that site access is available in order to identify the need for SI for any additional hot spots as a result of the on-going land contaminating activities. In addition, re-appraisal would be required for the other surveyed sites, other remaining areas of the PDA and the works areas for the associated infrastructure to address any change in land use that may give rise to potential land contamination issues.

16.10.1.7 Findings from the re-appraisal will be presented in a supplementary CAP. Upon approval of the supplementary CAP and completion of the SI works, a CAR would be prepared to present findings of the SI works. If contamination has been identified, a RAP would be prepared to recommend specific remediation measures. Upon completion of the remediation works, if any, a RR would also be prepared to demonstrate that the clean-up works are adequate. The CAR, RAP and RR would be submitted to EPD for approval prior to commencement of any construction / development works.

16.10.1.8 Upon remediation of the contamination land, if any, the Project will have converted previously potentially contaminated soil and groundwater into safe, usable land fit for development, thus bringing benefits to the community at large and helping to address Hong Kong's long-term housing demand and other land use needs.

16.11 Cultural Heritage

16.11.1.1 A desk-top study has been conducted to identify cultural heritage resources within the PDA boundary and works boundary outside PDA. There are no Sites of Archaeological Interest within the PDA boundary and works boundary outside PDA. A terrestrial archaeological survey has been conducted on Government land in the TYST area. It was concluded that surveyed areas were undisturbed until recently, and that the part of the area in the north has low archaeological potential. For the private lands that have "Moderate" archaeological potential within the PDA, they are constrained by accessibility issues and hence surveys cannot be conducted during this EIA. In the previous surveyed government land area, part of the "Southern Study Area" in Area 1 adjacent the location of Song/Qing cloth impressed tiles has been subject to an archaeological review which has recommended that this area be subject to archaeological survey prior to construction. Subject to the findings of

archaeological work, appropriate mitigation measures would be proposed by the project proponent in prior agreement with the AMO before the construction phase of the proposed development. Further archaeological survey will be conducted in the concerned area after land resumption. The areas proposed for archaeological survey represent areas of moderate archaeological potential to be impacted by development across interpreted natural or undisturbed ground.

16.11.1.2 The scope and programme of the proposed archaeological work will be agreed with AMO. Subject to the findings of the archaeological work, appropriate mitigation measures would be proposed by the Project proponent in prior agreement with the AMO before the construction phase of the proposed development.

16.11.1.3 The impacts of the new road alignments in TYST and TSWW Interchange, new sewer and Reclaimed Water Service Reservoir under the RODP will not require archaeological assessment. These proposed roads and sewer network are located at a highly disturbed and urbanised area, and the proposed Reclaimed Water Service Reservoir is situated on an existing hill slope, it is predicted that the archaeological potential are likely low. These areas will not be surveyed after land resumption. There are some small areas to the west and south of the TYST Interchange which will be surveyed after land resumption.

16.11.1.4 Given the results in southern part of the surveyed Area 1, an archaeological review has recommended that this area be subject to archaeological survey prior to construction. Subject to the findings of archaeological work, appropriate mitigation measures would be proposed by the project proponent in prior agreement with the AMO before the construction phase of the proposed development. For the areas with low-moderate archaeological potential, AMO should be informed immediately in case of discovery of antiquities or supposed antiquities in the course of the construction works. Agreement from AMO would be sought on the follow-up actions if required.

16.11.1.5 A Grade 3 historic building, the Yeung Hau temple, lies within the PDA at TYST and should be preserved. There are two declared monuments (Cheung Ancestral Hall at Shan Ha Tsuen and Tang Ancestral Hall at Ha Tsuen) within 100m assessment area but outside PDA. In addition 14 graded, 1 proposed graded and 1 nil graded historic building are identified within 100m assessment area but outside PDA and these buildings will not be impacted by works.